

Robotic CyberKnife Radiosurgery for the Treatment of Choroidal and Ciliary Body Melanoma



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• **PURPOSE:** To present the results of robotic-assisted radiosurgery in choroidal and ciliary body melanomas treated at the Department of Ophthalmology of Ludwig-Maximilians-University and the European CyberKnife Center in Munich, Germany.

• **DESIGN:** Interventional case series

• **METHODS:** This retrospective study included 594 consecutive patients referred to our clinic for the treatment of choroidal and ciliary body melanomas with robotic radiosurgery (CyberKnife) from 2005 to 2019. Eye retention, local control and disease-specific survival rates were calculated as Kaplan–Meier and actuarial estimates. The impact of prescription dose, tumor size, and ciliary body involvement was assessed by likelihood ratio tests and Cox regression.

• **RESULTS:** Among all patients who were staged according to the TNM classification system (8th edition), 22.7% were I, 57.9% were II, 18.9% were III, and 0.5% were IV. Median apical tumor height and base diameter were 5.8 and 11.4 mm. The mean follow-up was 41.7 months. Local control after 3 and 5 years was 92.0% (95% CI = 88.2%-94.7%) and 84.3% (95% CI = 77.9%-89.0%), respectively, for 21 to 22 Gy and 86.9% (95% CI = 79.7%-91.7%) and 77.7% (95% CI = 68.5%-84.6%), respectively, when treated with 20 Gy or less. Eye retention was achieved in 89.9% and 81.0% after 3 and 5 years with 21 to 22 Gy and 85.9% and 80.0% for 20 Gy or less. Disease-specific survival rates were 93.1% (95% CI = 90.2%-95.2%) after 3 years, 89.8% (95% CI = 86.0%-92.6%) after 5 years, and 87.8% (95% CI = 82.8%-91.4%) after 7 years.

• **CONCLUSIONS:** This is the largest series of patients treated for choroidal and ciliary body melanomas with CyberKnife. Our results reflect an improvement in the outcome of CyberKnife therapy for patients with choroidal

and ciliary body melanoma treated with single-session radiosurgery in the last decade. (Am J Ophthalmol 2023;250: 177–185. © 2022 Published by Elsevier Inc.)

UVEAL MELANOMA IS ALMOST EXCLUSIVELY TREATED with some form of irradiation if enucleation can be avoided. Brachytherapy has emerged to become the most widely used treatment option in choroidal melanoma, and the Collaborative Ocular Melanoma Study Group (COMS) has extensively contributed to our understanding and knowledge in the management of uveal melanoma with plaque radiotherapy.^{1,2} However, restrictions such as tumor size and tumor location exist and limit the use of brachytherapy in uveal melanoma. Ruthenium-106 (Ru-106) plaques, for example, are usually considered in melanomas with an apical height not exceeding 7 mm. Tumors located directly beneath the fovea or in close proximity to the optic nerve are also not very suitable for treatment with radioactive plaques.^{3–5} External beam radiation (EBR) approaches have existed for almost as long as brachytherapy but are usually considered only in cases when brachytherapy is not a suitable treatment option. The linear accelerator (LINAC) is 1 such EBR device and is based on the generation of photons through the acceleration of charged particles. It is also the most commonly used EBR option in the treatment of patients with any form of cancer and is thus available in most major hospitals throughout the world. Treatment of uveal melanoma using a LINAC is a proven treatment option, and several reports have documented similar outcomes regarding local control and overall survival in comparison with the standard of care, proton beam therapy (PBT), in uveal melanoma.^{6–9} In addition, the spectrum and frequency of complications following radiation therapy using photons is also similar to PBT.^{10–13}

When treatment with a LINAC is carried out with a single fraction, it is usually referred to as stereotactic radiosurgery (SRS), a widely used and available method for the treatment of tumors in all parts of the body including brain tumors and metastases.^{14,15} Stereotactic radiotherapy (SRT) uses the same equipment and technical devices but is performed over several days and usually uses smaller radiation fractions per session. It is unclear, however, whether a single radiation session with a higher radiation dose as in

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